Pietro Adamvoski

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EDUCATION

Carleton University Ottawa, ON
Bachelor of Software Engineering | Graduated July 2026

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Java, Python, SQL, Go, C, Racket, HTML/CSS
Frameworks: React, Next.js, Angular, Node.js, NestJS, Express, Spring Boot, FastAPI
Data and infra: PostgreSQL, pgvector, Prisma, Redis, SQLite, Docker, Terraform, AWS, GitHub Actions
Tools and testing: Git, Arch Linux, OpenCode, JUnit, Testcontainers, pytest, Vitest, Playwright, Maven

PROJECTS

SupportLens | AI helpdesk with cited answers

Python, FastAPI, React, TypeScript, PostgreSQL, pgvector, Celery, Redis, Docker, Claude API

- Some support docs are restricted, so an AI reply could quote one to the wrong agent or make unchecked claims.
- Built a helpdesk where Claude drafts ticket replies from a knowledge base and cites a source passage for each claim, fed by a Celery pipeline that chunks, embeds and versions uploaded docs.
- Put the permission filter inside the hybrid search SQL (pgvector plus Postgres full-text, fused with RRF) so it runs before ranking and restricted chunks never reach the prompt.
- Measured 28 of 35 correct answer-or-refuse decisions in an offline eval run, and confirmed that removing the permission filter fails 10 of 12 retrieval tests.

FulfillOps | Order and inventory operations console

Java 21, Spring Boot, PostgreSQL, Angular, Testcontainers, Playwright, Terraform, AWS ECS

- Two reps can sell the last unit of a product at once, and a retried carrier request can book the same parcel twice.
- Built an order and inventory console with stock reservations, an order state machine and a stock ledger that a Postgres trigger keeps append-only.
- Replaced check-then-act stock logic with one conditional UPDATE backed by a CHECK constraint, and saved each carrier idempotency key before the first attempt so retries return the original booking.
- Tested the race with a mutation check, where the check-then-act version failed 48 of 50 concurrent races, and wrote 176 backend tests against real PostgreSQL in Testcontainers.

ScopeFlow | Multi-tenant SaaS for agency scope changes

TypeScript, Next.js, NestJS, PostgreSQL, Prisma, WebSockets, Playwright

- Agencies lose money on scope creep when they can’t prove which version of a change a client approved, and two people editing one proposal can overwrite each other.
- Built a multi-tenant app where proposals are stored as immutable revisions, approvals point at the exact revision the client saw, and stale edits get a 409 with a field-by-field diff.
- Enforced tenant isolation in Postgres with row-level security set per transaction, so a query that forgets its org filter still can’t read another org’s rows.
- Tested isolation by sending another org’s IDs to 21 API routes; with row-level security bypassed, 9 of 21 requests read or changed that org’s data.

Smart Scheduler | Carleton engineering capstone

TypeScript, React, Express, PostgreSQL, Prisma

- Carleton students plan terms by hand, cross-checking their audit, prerequisites and the timetable for clashes.
- Built a course planner that reads a student’s audit, checks prerequisites and generates all possible combinations of conflict-free timetables for the courses they still need.
- Stored prerequisite rules as trees in PostgreSQL to handle nested AND/OR conditions, minimum grades and year standing, and evaluated them recursively against a student’s record.
- Parsed prerequisite trees for 322 of 348 catalog courses and wrote 107 unit tests, including 16 simulated audits compared against the histories they were generated from.

EXPERIENCE

Electronics Repair Technician | uBreakiFix (now Mobile Klinik) June 2020–September 2021
Kanata, ON

- Diagnosed and repaired hardware, software and connectivity faults in phones, laptops and other electronic devices.
- Tested each repaired device to confirm the original fault was fixed before returning it to the customer.
- Ordered replacement parts, managed parts inventory and processed warranty repairs under protection-plan rules.